

Commerce XI Maths App

Theory (Pages 8-15)

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 1 - Partition Values | Total questions: 42

Question 1 - MCQ · PG-8 · TH-Q1

Deciles divide ordered observations into

Options

- A. 10 equal parts
- B. 4 equal parts
- C. 2 equal parts
- D. 100 equal parts

Correct answer: 10 equal parts

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 10 equal parts.

Question 2 - MCQ · PG-8 · TH-Q2

The number of deciles is

Options

- A. 9
- B. 10
- C. 4
- D. 99

Correct answer: 9

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 9.

Question 3 - MCQ · PG-8 · TH-Q3

Deciles are denoted by

Options

- A. $D_1, D_2, \dots, D_9$
- B. Q_1, Q_2, Q_3
- C. $P_1, P_2, \dots, P_{99}$
- D. $D_0, D_1, \dots, D_{10}$

Correct answer: $D_1, D_2, \dots, D_9$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $D_1, D_2, \dots, D_9$.

Question 4 - MCQ · PG-8 · TH-Q4

For raw data, the position of D_i is

Options

- A. $(i(n+1))/(10)$
- B. $(i(n+1))/(4)$
- C. $(iN)/(100)$
- D. $((n+i))/(10)$

Correct answer: $(i(n+1))/(10)$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(i(n+1))/(10)$.

Question 5 - MCQ · PG-8 · TH-Q5

For grouped data, the decile class contains the

Options

- A. $((iN)/(10))^{\text{th}}$ observation
- B. $((iN)/(4))^{\text{th}}$ observation
- C. $((iN)/(100))^{\text{th}}$ observation
- D. N^{th} observation

Correct answer: $((iN)/(10))^{\text{th}}$ observation

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $((iN)/(10))^{\text{th}}$ observation.

Question 6 - MCQ · PG-8 · TH-Q6

Percentiles divide ordered observations into

Options

- A. 100 equal parts
- B. 10 equal parts
- C. 4 equal parts
- D. 2 equal parts

Correct answer: 100 equal parts

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 100 equal parts.

Question 7 - MCQ · PG-8 · TH-Q7

The number of percentiles is

Options

- A. 99
- B. 100
- C. 9
- D. 3

Correct answer: 99

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 99.

Question 8 - MCQ · PG-8 · TH-Q8

Percentiles are denoted by

Options

- A. $P_1, P_2, \dots, P_{99}$
- B. $D_1, D_2, \dots, D_9$
- C. Q_1, Q_2, Q_3
- D. $P_0, P_1, \dots, P_{100}$

Correct answer: $P_1, P_2, \dots, P_{99}$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $P_1, P_2, \dots, P_{99}$.

Question 9 - MCQ · PG-8–9 · TH-Q9

For raw data, the position of P_i is

Options

- A. $(i (n + 1))/(100)$
- B. $(i (n + 1))/(10)$
- C. $(i N)/(4)$
- D. $((n + i))/(100)$

Correct answer: $(i (n + 1))/(100)$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(i (n + 1))/(100)$.

Question 10 - MCQ · PG-9 · TH-Q10

For grouped data, the percentile class contains the

Options

- A. $(i N)/(100)$ th observation
- B. $(i N)/(10)$ th observation
- C. $(i N)/(4)$ th observation
- D. N^{th} observation

Correct answer: $(i N)/(100)$ th observation

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(i N)/(100)$ th observation.

Question 11 - MCQ · PG-9 · TH-Q11

Which relation is correct?

Options

- A. $Q_2 = D_5 = P_{50}$
- B. $Q_1 = D_5 = P_{50}$
- C. $Q_3 = D_5 = P_{50}$
- D. $Q_2 = D_2 = P_{20}$

Correct answer: $Q_2 = D_5 = P_{50}$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $Q_2 = D_5 = P_{50}$.

Question 12 - MCQ · PG-9 · TH-Q12

Which relation is correct?

Options

- A. $Q_1 = P_{25}$
- B. $Q_1 = P_{50}$
- C. $Q_2 = P_{25}$
- D. $Q_3 = P_{25}$

Correct answer: $Q_1 = P_{25}$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $Q_1 = P_{25}$.

Question 13 - MCQ · PG-9 · TH-Q13

Which relation is correct?

Options

A. $Q_3 = P_{75}$

B. $Q_3 = P_{25}$

C. $Q_2 = P_{75}$

D. $Q_1 = P_{75}$

Correct answer: $Q_3 = P_{75}$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $Q_3 = P_{75}$.

Question 14 - MCQ · PG-9 · TH-Q14

D_1 is equivalent to

Options

A. P_{10}

B. P_1

C. P_{25}

D. P_{50}

Correct answer: P_{10}

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives P_{10} .

Question 15 - MCQ · PG-9 · TH-Q15

P_{30} is equivalent to

Options

A. D_3

B. D_1

C. D_5

D. D_7

Correct answer: D_3

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives D_3 .

Question 16 - MCQ · PG-9 · TH-Q16

D_7 is equivalent to

Options

A. P_{70}

B. P_7

C. P_{35}

D. P_{75}

Correct answer: P_{70}

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives P_{70} .

Question 17 - MCQ · PG-10 · TH-Q17

D_4 identifies the value below which approximately what percentage of observations lie?

Options

A. 40%

B. 4%

C. 25%

D. 60%

Correct answer: 40%

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 40%.

Question 18 - MCQ · PG-10 · TH-Q18

P_{21} identifies the value below which approximately what percentage of observations lie?

Options

A. 21%

B. 79%

C. 2.1%

D. 50%

Correct answer: 21%

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 21%.

Question 19 - MCQ · PG-11 · TH-Q19

For grouped data, N represents

Options

- A. total frequency
- B. class frequency only
- C. class width
- D. lower boundary

Correct answer: total frequency

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives total frequency.

Question 20 - MCQ · PG-11 · TH-Q20

The class containing $(i N)/(10)$ is called the

Options

- A. i^{th} decile class
- B. i^{th} quartile class
- C. median class only
- D. percentile class

Correct answer: i^{th} decile class

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives i^{th} decile class.

Question 21 - MCQ · PG-11 · TH-Q21

The class containing $(i N)/(100)$ is called the

Options

- A. i^{th} percentile class
- B. i^{th} decile class
- C. quartile class
- D. modal class

Correct answer: i^{th} percentile class

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives i^{th} percentile class.

Question 22 - MCQ · PG-12 · TH-Q22

A missing frequency can be found from a known partition value by

Options

- A. substituting it in the appropriate partition formula
- B. using only the range
- C. ignoring cumulative frequency
- D. using the class mark as N

Correct answer: substituting it in the appropriate partition formula

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives substituting it in the appropriate partition formula.

Question 23 - MCQ · PG-13 · TH-Q23

If data are given by mid-values, class boundaries are obtained using

Options

- A. half the difference between consecutive mid-values
- B. the total frequency
- C. the largest frequency
- D. the median only

Correct answer: half the difference between consecutive mid-values

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives half the difference between consecutive mid-values.

Question 24 - MCQ · PG-14 · TH-Q24

The limits of the middle 60% of observations are

Options

- A. P_{20} and P_{80}
- B. P_{30} and P_{70}
- C. P_{40} and P_{60}
- D. P_{10} and P_{90}

Correct answer: P_{20} and P_{80}

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives P_{20} and P_{80} .

Question 25 - MCQ · PG-14 · TH-Q25

The limits of the middle 40% of observations are

Options

A. P_{30} and P_{70}

B. P_{20} and P_{80}

C. P_{40} and P_{60}

D. Q_1 and Q_3

Correct answer: P_{30} and P_{70}

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives P_{30} and P_{70} .

Question 26 - Enter answer · PG-9 · TH-Q26

For $n = 10$, the position of D_3 isth observation

Correct answer: 3.3

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 3.3.

Question 27 - Enter answer · PG-9 · TH-Q27

For $n = 10$, the position of P_{70} isth observation

Correct answer: 7.7

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 7.7.

Question 28 - Enter answer · PG-10 · TH-Q28

For $n = 95$, the position of D_4 isth observation

Correct answer: 38.4

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 38.4.

Question 29 - Enter answer · PG-10 · TH-Q29

For $n = 95$, the position of P_{55} isth observation

Correct answer: 52.8

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 52.8.

Question 30 - Enter answer · PG-11 · TH-Q30

For $n = 15$, the position of D_6 isth observation

Correct answer: 9.6

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 9.6.

Question 31 - Enter answer · PG-11 · TH-Q31

For $n = 15$, the position of P_{85} isth observation

Correct answer: 13.6

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 13.6.

Question 32 - Enter answer · PG-9 · TH-Q32

For the ordered data :

169, 225, 289, 324, 325, 400, 625, 729, 784, 841.

Calculate D_3

Correct answer: 299.5

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 299.5.

Question 33 - Enter answer · PG-9 · TH-Q33

For the ordered data :

169, 225, 289, 324, 325, 400, 625, 729, 784, 841.

Calculate P_{70}

Correct answer: 697.8

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 697.8.

Question 34 - Enter answer · PG-10 · TH-Q34

No. of defective products	30	35	40	45	50	55	60
No. of firms	12	35	10	15	8	7	8

Calculate D_4 .

Correct answer: 35

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 35.

Question 35 - Enter answer · PG-10 · TH-Q35

No. of defective products	30	35	40	45	50	55	60
No. of firms	12	35	10	15	8	7	8

Calculate P_{55} .

Correct answer: 40

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 40.

Question 36 - Enter answer · PG-10–11 · TH-Q36

Profit (in ₹ lakh)	0.5–4.5	5.5–9.5	10.5–14.5	15.5–19.5	20.5–24.5
No. of firms	7	18	25	30	20

Calculate D_4 .

Correct answer: 13

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 13.

Question 37 - Enter answer · PG-10–11 · TH-Q37

Profit (in ₹ lakh)	0.5–4.5	5.5–9.5	10.5–14.5	15.5–19.5	20.5–24.5
No. of firms	7	18	25	30	20

Calculate P_{21} .

Correct answer: 8.89

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 8.89.

Question 38 - MCQ · PG-14 · TH-Q38

If 75% of observations lie below a value, that value is

Options

A. Q_3 or P_{75}

B. Q_1 or P_{25}

C. Q_2 or P_{50}

D. D_3 or P_{30}

Correct answer: Q_3 or P_{75}

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives Q_3 or P_{75} .

Question 39 - MCQ · PG-14 · TH-Q39

If 20% of observations lie below a value, that value is

Options

- A. P_{20} or D_2
- B. P_{80} or D_8
- C. Q_1
- D. Q_2

Correct answer: P_{20} or D_2

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives P_{20} or D_2 .

Question 40 - MCQ · PG-15 · TH-Q40

When the desired partition position falls in a class, interpolation uses the frequency of

Options

- A. that partition class
- B. the first class
- C. the last class
- D. the modal class only

Correct answer: that partition class

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives that partition class.

Question 41 - Enter answer · PG-15 · TH-Q41

Delivery time (minutes)	0-10	10-20	20-30	30-40	40-50
Number of orders	12	18	30	25	15

Find the percentage of orders delivered in less than 25 minutes.

Correct answer: 45%

Explanation and steps

The classes are already continuous. Let $P_i = 25$, where i is the required percentage. The percentile class is 20-30, so $L = 20$, $h = 10$, $f = 30$, $c.f. = 12 + 18 = 30$ and $N = 100$. Using $P_i = L + (h/f)(iN/100 - c.f.)$:

$$25 = 20 + (10/30)(i - 30)$$

$$15 = i - 30$$

$$i = 45$$

Therefore, 45% of the orders were delivered in less than 25 minutes.

Question 42 - Enter answer · PG-15 · TH-Q42

Marks	10-19	20-29	30-39	40-49	50-59
Number of students	8	12	40	25	15

Find the percentage of students who obtained less than 35 marks.

Correct answer: 42%

Explanation and steps

The inclusive classes must first be made continuous: 9.5-19.5, 19.5-29.5, 29.5-39.5, 39.5-49.5 and 49.5-59.5. Let $P_i = 35$, where i is the required percentage. The percentile class is 29.5-39.5, so $L = 29.5$, $h = 10$, $f = 40$, $c.f. = 8 + 12 = 20$ and $N = 100$. Using $P_i = L + (h/f)(iN/100 - c.f.)$:

$$35 = 29.5 + (10/40)(i - 20)$$

$$22 = i - 20$$

$$i = 42$$

Therefore, 42% of the students obtained less than 35 marks.